
Report: Measuring the Internet

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I Introduction

In recent years, the Internet has become essential to our daily lives. We use it everywhere, always... Without even knowing what it is made of, we use it to access websites, make shopping, get information, make phone calls... However, behind those services lies a huge communication network composed of billions of elements linked to each other.

The aim of this project was thus above all to understand the functioning of this network and to search on the way data were transmitted over the Internet. Then, a second part of it, in parallel, was to answer the question “What is the best way to measure the Internet?”. This subject was very large and we struggled to understand its meaning. However, after some time, some research and some work on our project, we concluded that the question led us to search, understand and test the different ways we had to “measure the Internet”. What we understood by this expression was the fact to get the “distance” between two hosts by measuring different quantities related to the Internet and the way it works.

Therefore, we divided each other the work : some of us worked on the more theoretical parts, making searches on the Internet to have a better understanding on how the Internet works. Some others concentrated on the experiments we did to measure the different quantities mentioned earlier, making graphs, trying to solve the different technical problems we encountered, and making conclusions out of what we obtained.

II Theory and definitions

First and foremost, we had to understand what was the Internet and what did we mean by this word. We needed to apprehend the way it worked so that we could well carry out our project. For that, some of us started by making a lot of searches and made a summary for the others so that we were all able to understand what we were doing in that project.

1 A network of networks

The Internet actually stands for “Interconnected networks”. Indeed, it is made of a multitude of little networks connected to each other. Let’s decompose this giant structure.

A network is a group of connected computers that are able to send data to each other, mainly via cables. Each device connected to a network is called a host. It can work as a server, offering information resources, services or applications ; as a client, making requests to other devices; or both.

However, we realize very quickly that it is absolutely not possible to connect each computer of the Internet to each other. Indeed, for only 20 computers, it already requires 190 connections. Therefore, we need to introduce routers, to which multiple computers are connected. All the computers attached to the same router form a little network. Then, by connecting routers the same way, we can finally obtain a giant network of smaller networks. This way, we have what

we call LANs, Local Area Networks, which correspond approximately to the size of a house or a building. Then, by connecting the LANs together, we get a WAN, corresponding to the size of a city, and finally, by connecting them together, we obtain the Internet.

2 IP address

To be able to communicate together, devices on a network need to be able to identify each other. For that purpose, each computer on the Internet is uniquely identified with an IP address.

The most commonly used type of IP address is the IPv4. Each address is then composed of 4 numbers between 0 and 255, which corresponds to 32 bits. This allows to create around 4 billion of IP addresses using this system. However, nowadays, there are way more devices than that connected to the Internet.

Therefore, we created IPv6. The IP addresses are much larger, composed of 120 bits, making possible the connection of more computers. This version is developing more and more but a majority of devices still uses IPv4. To manage the lack of number of IP addresses, these computers use private addresses. This way, the IP doesn't need to be unique on the global internet but just on a local network. When a private address makes a request, it passes through its local network's router which makes the request with its own public IP. When the request is sent back to the router, the latter forwards it to the good private IP.

3 Data sending

Now that we have seen what a network is made of, we are going to see how the computers, once connected, can exchange data.

First, the data are sent between computers in the form of bits, the most basic unit of information. According to the medium through which data are passing, bits can be translated into light or electric pulses.

Then, the data are gathered into what we call packets, a formatted unit of data carried by a packet-switched network. For example, if a user needs to load an image, the image file does not go from the web server to the user's computer in one piece. Instead, they are broken down into packets of data, and then reassembled by the user's computer into the original photo. Packets are made of a header and a payload. Data in the header are only used by networking hardware to direct the packet to its destination: it can be network addresses, error detections, codes... The payload is the actual message that will be extracted and then used.

Once this is done, the packets travel to the right destination, identified by an IP address. As said earlier, the computers can't all be directly connected. Therefore, during their path, the packets pass through a lot of different devices. Each step is what we call a hop.

III

Measuring the Internet

Once we understood better the Internet, we went to the next part, which consisted of trying to measure by different ways the “distance” between two hosts on the Internet. For this, we needed to first apprehend the tools we were going to use : ping and traceroute. Some of us made searches on them and tested them on random hosts. Then, once again, they made a summary for the others to be comfortable with the tools. Some others then started to study the different ways to measure the Internet, tracing graphs and making maps using python. After that, we all discussed about the results obtained to give them an interpretation. We sometimes also needed to make searches to understand what happened, in which case we were splitting the work. Therefore, in a first part, we will estimate the time that separates two devices. Then, we will measure the distance travelled by the packets to go from a host to another, the number of hops on the path, and finally the physical distance that separates the computers.

1

Measurement of the time separating two hosts

a

Estimation of the propagation time with "Ping"

The time taken by data to travel from a host to another, also called propagation time, can be measured thanks to a tool mainly used to diagnose network issues called “Ping”. Actually, this tool enables us to know the time taken by packets to perform a round-trip to a given host on the Internet.

In the terminal, you type the command ping followed by the IP address of the server you want to test your connectivity with. The computer sends several packets to the given host and measures the time between the moment it sends them and the moment it gets them back. If it doesn't get it back after some point, then the packet is considered as lost.

In the example below, we decided to ping google.com. On the terminal, we write `ping google.com`. Then, we can see that the terminal displays on the second line the IP address corresponding to the address given, here 142.250.179.78. Then, each line represents the sending of a packet whose number is indicated by the information `icmp_seq`. The information `ttl` means “Time To Live”, it's the maximum number of hops the packet can take before to be discarded. Here, the most important information for us is the time displayed at the end of each line. Finally, at the bottom of the ping command are written some statistics : the percentage of packets lost and `min/avg/max/mdev` which correspond respectively to minimum, average, maximum and mean deviation of the times.

```
$ ping google.com
PING google.com (142.250.179.78) 56(84) bytes of data.
64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=1
    ttl=110 time=12.7 ms
```

```

64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=2
  _ ttl=110 time=28.9 ms
64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=3
  _ ttl=110 time=15.7 ms
64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=4
  _ ttl=110 time=32.2 ms
64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=5
  _ ttl=110 time=16.2 ms
64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=6
  _ ttl=110 time=15.5 ms
64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=7
  _ ttl=110 time=15.7 ms
64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=8
  _ ttl=110 time=13.7 ms
64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=9
  _ ttl=110 time=15.6 ms
64 bytes from par21s19-in-f14.1e100.net (142.250.179.78): icmp_seq=10
  _ ttl=110 time=16.2 ms
--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9015 ms
rtt min/avg/max/mdev = 12.685/18.243/32.217/6.287 ms

```

Figure III.1: Example of a ping command display for google.com

To be able to do that, ping uses the ICMP (Internet Control Message Protocol). ICMP is a network layer protocol used by network devices to diagnose connectivity issues. The primary purpose is error reporting. When 2 devices connect over the Internet, ICMP generates errors to share with the sending device if any of the data did not reach its intended destination. More generally, it is used to follow the emission of a packet and report any dysfunction. When we use a ping, it generates an ICMP packet with the message “Echo request” that corresponds to a certain type and a certain code in the ICMP packet. That way, the destination server, when it receives it, knows it has to send it back with the message “Echo reply”.

The main aims of the ping are to check the communication between two devices connected to a network by verifying if the packets are well sent back, to measure the reliability of a network at instant t by looking at the loss percentage, and to measure the latency of a network at instant t .

However, the round-trip time obtained with the ping is actually composed of different delays and not only the propagation one. The main ones are :

- The Processing Delay (D_p) : it's a delay due to the processing of packet headers by routers. The delay is due to checking bit errors, figuring out the next hop to which

packets need to be sent, and encryption operations.

- Queuing delay (D_Q) : it's the time taken in routing queues. Queuing delay is the time data waits in the buffer of a router. If there is a congestion of the network, it can represent a significant time.
- Transmission delay (D_T) is the time to push all the available data in a transmission medium or wire. $D_T = \frac{N_B}{T_R}$ with N_B the number of bits and T_R the transmission rate.
- Propagation delay (D_{PR}) is the one we want to estimate. It's the time data actually spend in the medium of transmission travelling from routers to routers. $D_{PR} = \frac{d}{v}$ with d the distance actually travelled by the data and v the speed at which data are transmitted.

The network latency is defined as the sum of all these delays. However, we must be cautious about the fact that round-trip time is most often not equal to twice the network latency. Indeed, the latency can be asymmetric and there is an extra processing time at the destination that has to be taken into account.

Knowing this, we can say the best estimation of the propagation delay thanks to the ping tool is the minimum because it represents the time when all the other delays are at their lowest. Indeed, when there is congestion of the network for example, the other delays distort the round-trip time such that we have no longer access to the propagation time. Therefore, the closest value to the propagation time is the minimum, although it's still the propagation time corresponding to the data travelling back and forth between the two hosts.

Due to many of the hosts dropping ICMP support, we had to use TCP (Transmission Control Protocol) to ping them. TCP is more reliable, but unfortunately, it isn't supported by Linux ping tool by default, so we had to write a simple Python function to ping using `mtrpacket`, which allows TCP pings.

```
import mtrpacket

async def ping(url):
    async with mtrpacket.MtrPacket() as mtr:
        res = await mtr.probe(url, protocol="tcp")
    return res
```

However, pinging once is never accurate, since there could be some unusual problems, we had to do 5 pings for each host and take the minimum among all the pings to calculate the actual latency.

```
async def min_ping(url):
    res = []
    for i in range(0, 5):
        p = await ping(url)
```

```

if hasattr(p, "time_ms") and p.time_ms is not None:
    res.append(p.time_ms)

return min(res)

```

Using Python also allows us to quickly ping many hosts at the same time and easily graph various things related to pings.

b Relationship between the time and the physical distance

We now can give an approximation of the propagation time thanks to ping but we could question the relevance of this measurement. Indeed, our first thought was that the round-trip time was tightly linked to the physical distance separating the two hosts.

To analyze the relationship between the two quantities, we wanted to plot a graph of the round-trip time as a function of the physical distance. For that, we made a lot of pings with different hosts from various locations. We computed the physical distance separating our computer to the hosts and obtained the graph below. The points are very scattered and don't really look like a straight line. However, as expected, it can be seen that physical distance affects the ping. Thus, we were able to approximate a straight line that we will use later on even if it's very inaccurate. Nevertheless, the time is still not completely linked to the physical distance : there are other parameters to take into account. Consequently, time appears as an independent measurement that can be relevant to do.

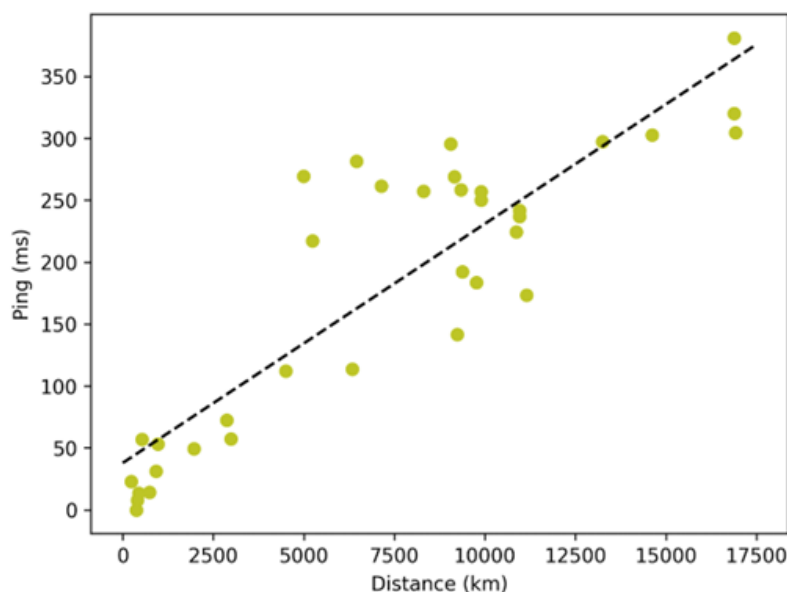


Figure III.2: Graph of the Round-Trip Time as a function of physical distance between two hosts

As another result, we knew that other parameters came into play so we tried to explain them. As said before, the propagation time depends on the distance travelled by packets and the velocity to which they travel. Let's see how these quantities can be influenced so that time varies that much.

c Medium of propagation of data

The first parameter we are going to dive on is the medium of propagation which greatly influences the speed of propagation. Let's detail the different ways to connect computers together:

The first one is by cables, category that can be split into two : copper cables and fiber optic cables.

Copper cables are the most commonly used. They are made from copper wires and use electrical pulses to transmit data. Each bit is then translated into one pulse. On the other hand, fiber optic cables have a silica glass fiber core. They transmit data via light pulses. They are what we call a more "premium" solution.

Indeed, fiber optics cables offer a faster transmission of data since light is faster than electrons. Moreover, they support greater cable length because they have a lower rate of loss of the signal. They are more durable and better resist the pressure and the stress. The communications are also safer because they are highly resistant to tapping, radiofrequency and jamming.

Nevertheless, we still have a majority of copper cables because they are much cheaper than fiber optic, even if it can be questioned because of the difference in durability. However, fiber optic cables have developed a lot in the last couple of years.

Another way to connect computers together is using satellites. Satellite internet utilizes communications satellites to provide internet access. It transmits data between a satellite dish and an orbiting satellite which allows people to connect to the internet regardless of their location. Indeed, cable networks are very developed in cities but in some remote places, it can be hard to get access to the Internet. From this point of view, satellites can be a good solution for areas where traditional wires may not be available such as planes, Antarctica or countryside.

However, it still doesn't offer the same quality as cables, explaining why not everyone uses them. Indeed, cables have a higher download and upload speed. On the contrary, satellite Internet, although it has improved in recent years, may experience latency issues due to the long distance data has to travel. Satellites are also more difficult to install. In addition to that, they are much more expensive even though they are less reliable than cables. This is due to the fact that they can be affected by weather conditions. It can cause signal interruptions or decreased performance. Finally, cables may be more secure since they don't involve signal transmission over open airwaves like satellite signals.

Actually, there are two types of satellites : according to it, the latency is not the same.

- Geostationary satellites : They are the oldest ones. These satellites are placed around 36 000 km above the Earth. Therefore, we can compute the minimum time it takes for data to be sent from the Earth to the satellite and to come back. At minimum, the distance between a computer and the satellite is 36 000 km, which makes 72 000 km back and forth. As electromagnetic waves travel to the speed of light, this makes a minimum of 240 ms of latency.
- Low orbit satellites : With the arrival of Starlink, created by Elon Musk, low orbit satellites start to play an important role in communications. Starlink is a large constellation of low-orbit satellites which are placed at around 550 km above Earth which is much closer, greatly decreasing the time for the data to travel from Earth to the satellites. They currently propose a latency of 50 ms. The long-term plan is to deploy over 12,000 internet satellites and to lower the latency to only 25 ms.

Since the satellites are much closer to Earth, they each cover a much lower surface on Earth, which explains why we need a “constellation” of satellites for a comprehensive coverage. To communicate between each other rapidly, each satellite is equipped with a laser communication terminal that uses optical inter-satellite links to transmit data. The use of lasers allows data to be transmitted at the speed of light which is faster in the vacuum than in fiber optic cables.

The big problem of this system is that it can pose a lot of challenges such as increased space debris, and potential interference with astronomical observations.

To conclude, we have seen here that there are many ways to connect computers together, having each advantages and drawbacks, and that they are each characterized by a proper speed of propagation. Therefore, for the same physical distance, according to the medium used, we can obtain two different values for the time.

d Physical distance doesn't reflect the distance travelled by packets

The other quantity involved in the expression of the time of propagation is the distance travelled by packets. Until there, we naively associated it to the physical distance but maybe they are not completely tied to each other. Indeed, as explained before, the path to go from a host to another is never straightforward, and the packets need to make a detour. Therefore, we will study more precisely how to measure this distance and use it as a measurement of the distance between two hosts.

2 Measurement of the distance travelled by packets

Just as the distance as crow flies between two places is not the distance we travel when doing the ride in car, the packets travel much more distance to go from a host to another than just the physical distance separating them. In order to compute this other distance, we need to know the path taken by data. For this, we need the tool named “Traceroute”.

The traceroute command is a network diagnostic tool that shows you the path your data takes from your computer to a destination on the Internet. It helps to identify the different devices through which the packets are passing.

On the terminal, we type the command traceroute followed by the IP address of the computer we want to reach. All the devices through which the data have passed in order to travel to it are then displayed. An example is provided below where we tried the traceroute with the address google.com once again. As with the ping, the first line of the terminal states the address given and the IP corresponding. The maximum number of hops along the path is also written. If the data need to make more stops, they will be automatically discarded after the maximum authorized and will never reach their destination. The next part of the display corresponds to the different devices encountered on the path. Each line describes a hop, giving the IP address of the device crossed as well as the time to travel to it.

```
$ traceroute google.com
traceroute to google.com (142.250.179.78), 30 hops max, 60 byte packets
 1 v152-rtr-doua2.rocad.fr (134.214.152.1) 0.353 ms 0.306 ms 0.318 ms
 2 10.100.254.3 (10.100.254.3) 0.458 ms 0.424 ms 0.451 ms
 3 10.11.3.10 (10.11.3.10) 1.332 ms 1.419 ms 1.478 ms
 4 10.11.3.1 (10.11.3.1) 1.234 ms 1.511 ms 1.462 ms
 5 193.55.215.134 (193.55.215.134) 2.042 ms 1.910 ms 1.599 ms
 6 vlan1845-be7-ren-nr-lyon1-rtr-091.noc.renater.fr (193.55.202.188)
   - 1.568 ms 1.584 ms 1.426 ms
 7 et-2-0-3-ren-nr-lyon1-rtr-131.noc.renater.fr (193.55.204.196) 7.748
   - ms 7.714 ms 7.893 ms
 8 xe-0-0-17-ren-nr-marseille1-rtr-131.noc.renater.fr (193.51.177.80)
   - 6.212 ms xe-0-0-8-ren-nr-marseille1-rtr-131.noc.renater.fr
   - (193.51.180.11) 6.428 ms xe-1-0-1-marseille1-rtr-131.noc.renater.fr
   - (193.51.177.222) 6.016 ms
 9 et-0-1-0-ren-nr-marseille2-rtr-131.noc.renater.fr (193.51.177.127)
   - 5.676 ms 6.066 ms 5.920 ms
10 72.14.218.132 (72.14.218.132) 6.264 ms 5.664 ms 5.825 ms
11 192.178.105.27 (192.178.105.27) 13.683 ms 192.178.105.209
   - (192.178.105.209) 13.305 ms 192.178.105.171 (192.178.105.171) 11.906
   - ms
12 192.178.105.80 (192.178.105.80) 132.317 ms 192.178.105.12
   - (192.178.105.12) 118.968 ms 119.010 ms
13 192.178.85.179 (192.178.85.179) 12.498 ms 72.14.234.191
   - (72.14.234.191) 16.561 ms 192.178.85.179 (192.178.85.179) 11.832 ms
14 64.233.175.243 (64.233.175.243) 12.482 ms * 12.324 ms
15 108.170.230.204 (108.170.230.204) 19.897 ms 209.85.248.116
   - (209.85.248.116) 12.901 ms 72.14.238.54 (72.14.238.54) 12.673 ms
```

```

16  108.170.255.169 (108.170.255.169)  13.745 ms  192.178.104.29
    ↳ (192.178.104.29)  12.102 ms  108.170.255.239 (108.170.255.239)  11.980
    ↳ ms
17  142.251.49.133 (142.251.49.133)  11.531 ms  11.591 ms  142.251.49.131
    ↳ (142.251.49.131)  11.494 ms
18  par21s19-in-f14.1e100.net (142.250.179.78)  11.268 ms  11.249 ms
    ↳ 11.519 ms

```

Figure III.3: Example of a traceroute command display for google.com

In order to be able to find out the path taken by data, Traceroute uses once again ICMP and the principle of the ping, as well as a very important parameter which is the Time To Live (TTL).

As explained before, the TTL is the maximum number of hops a packet can make before to be destroyed. Each router through which data pass decreases this number of one. Once it reaches 0, an ICMP error packet “Time to live exceeded” is sent back to the source. Therefore, as it can be seen on the image below, traceroute uses this principle and sends multiple packets with each time incrementing the TTL of one, starting from one. This way, each packet reveals a new device on the way, as this device will need to send an error message back. Moreover, to be able to know the time, traceroute uses the ping tool that allows to know the round-trip time from the source to each of the computers crossed. Once the packet arrives to its destination, the latter sends back an ICMP packet “Echo reply” : the traceroute is complete.

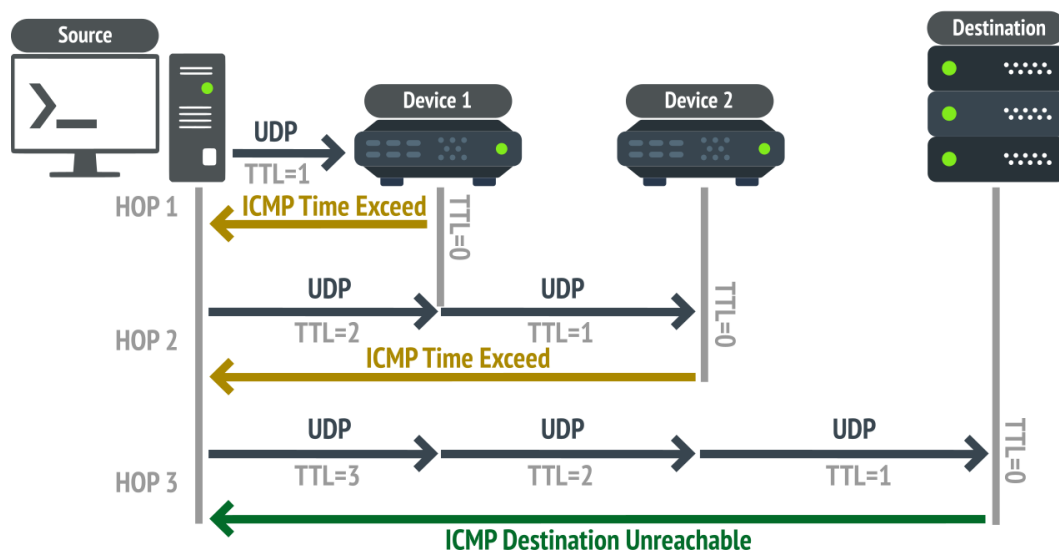


Figure III.4: Functioning of the traceroute tool

On several of our traceroute runs, we encountered asterisks on some hops and we wondered

what it meant. We then found out that the presence of an asterisk typically indicates that the router at that specific hop did not respond the traceroute program. There can be different reasons for that:

- First, a router may be too busy to respond to the traceroute probe, which it considers a low-priority event. We can check if that is the case by repeating traceroute for the same destination a few times. If we get the same list of routers but don't see "***" anymore, then it was a temporary overload. Similarly, if the path changes, that's probably because the router in question is under a heavy load, so it's temporarily avoided by other routers in the network.
- Another explanation is simply that some organizations configure their routers not to respond to traceroutes because they don't want to reveal details on their internal network. Then, a sequence of three-star hops that persists in repeated traceroute probes even if we increase TTL indicates that the routers don't respond because their owners forbade them to.

In the example given after, we see that we got responses up to step 15, but not beyond that. By repeating the experiment, we always ended up with the same result. We can thus conclude that the organization has set up its routers not to respond to traceroute, which gives up after 30 steps. We will not have any success even if we increase the maximum TTL, which we can do using "-m" in the initial command, since no router beyond the one in step 15 will respond.

```
$ traceroute -n mit.edu
traceroute to mit.edu (23.202.69.116), 30 hops max, 60 byte packets
...
14  23.32.63.13      76.969 ms  75.976 ms  74.613 ms
15  23.207.239.37    74.207 ms  74.402 ms  74.315 ms
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

Figure III.5: Example of an incomplete traceroute because of routers forbidden to answer

Again, we actually used TCP instead of ICMP to traceroute using Python. The library that we used for ping can also be used to traceroute, since it is capable of limiting the TTL.

```
async def trace(url):
    result = []

    async with mtrpacket.MtrPacket() as mtr:
        for ttl in range(1, 50):
            packet = await mtr.probe(url, protocol="tcp", ttl=ttl)

            if packet.responder:
                result.append(packet.responder)

            if packet.success:
                break

    return result
```

Using this function, we can traceroute many hosts very quickly:

```
functions = [trace(host) for host in hosts]
traces = await asyncio.gather(*functions)
```

Now that we have seen how to retrieve the path taken by data, we will be able to better understand what happens when a packet travels over the Internet. In that objective, we decided to plot the route taken by data for a lot of different traceroutes in order to highlight the highways of the Internet and to try to analyze the relationship between the physical distance and the actual distance travelled.

b Tracing of the map of the path taken by data

To create the following map, we chose plenty of destinations in very different places so that we could visualize better how the cables density and therefore distance travelled can change as a function of the locations. Once we chose a list of them, we decided to create a python code to automatize the creation of the map.

The result obtained allowed us to see that there are actually what we could call “highways” of the Internet, such as between the UK and the USA for example. This means that a lot of data pass through these routes to go to other places, other countries sometimes, which makes the distance much higher than what it could be. There are also some kind of hubs such as Paris in France for example. Indeed, we could see during our experiments that very often, no matter where we wanted to go in France or outside of it, the packets were always passing through Paris before to be redirected to their destination, which extended the distance once again.

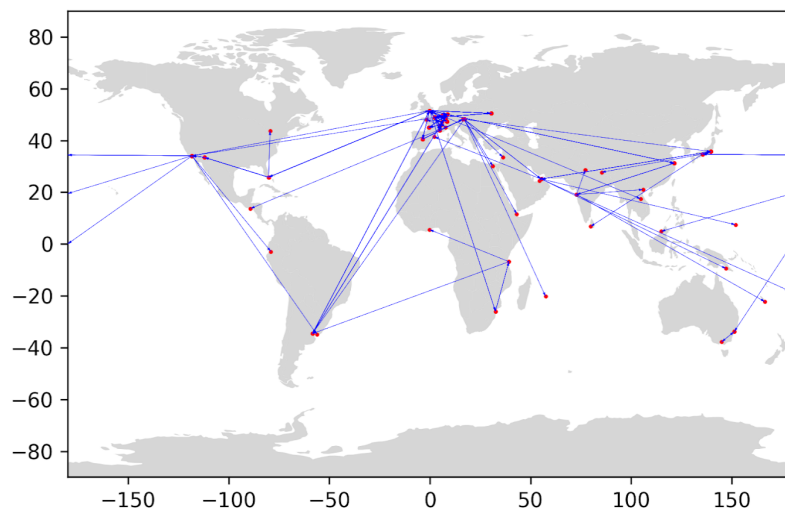


Figure III.6: Map of the path taken by data to different places all over the world

A striking example is given there. Instead of an expected straight line from France to the United Arab Emirates, the data made a complete turn around the world, passing as said before through Paris and using a highway from UK to the USA.

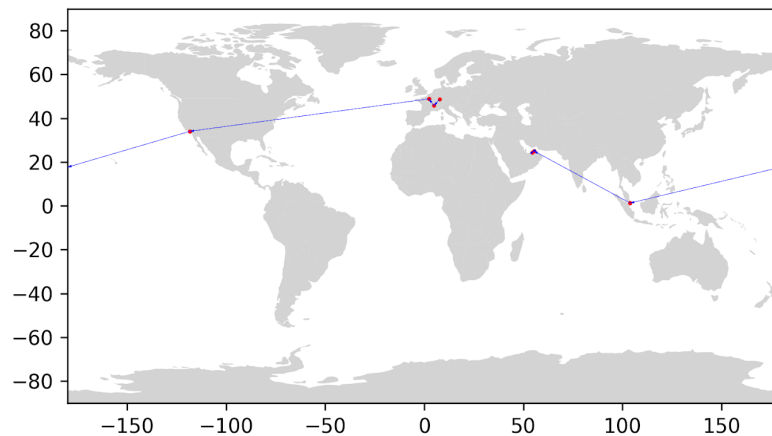


Figure III.7: Mapping of a traceroute to UAE

This phenomena actually happens regularly and can be explained by the fact that the connections go through the already available cables, which are not always straightforward between two specific locations. It explains why for a same physical distance, the ping can be so different. The actual distance travelled by the packets can indeed be much higher or not according to the links available between the two hosts. By measuring for each portion of the path the distance

corresponding and by making the sum, we could thus find a new way to measure the “distance” between two hosts.

In conclusion, this new quantity measured is quite relevant because it represents better the reality than physical distance for example. However, it is much less easy to measure : it takes a lot of time to make the traceroute and to find the distance between each host of the list obtained.

3 Measurement of the number of hops

Another measurement possible, much easier and faster, is to count the number of hops a packet makes between two devices. Once we have the traceroute, this is very simple to count the number of hops displayed on the terminal.

By repeating this process for multiple destinations, we very quickly realized it had no link with the physical distance. To verify our assumption and to visualize a bit better the possible link between the two, we plotted once again thanks to python the number of hops as a function of the physical distance and finally got the following graph. The latter supports perfectly our theory that there is no relationship between the two as the points seem completely random.

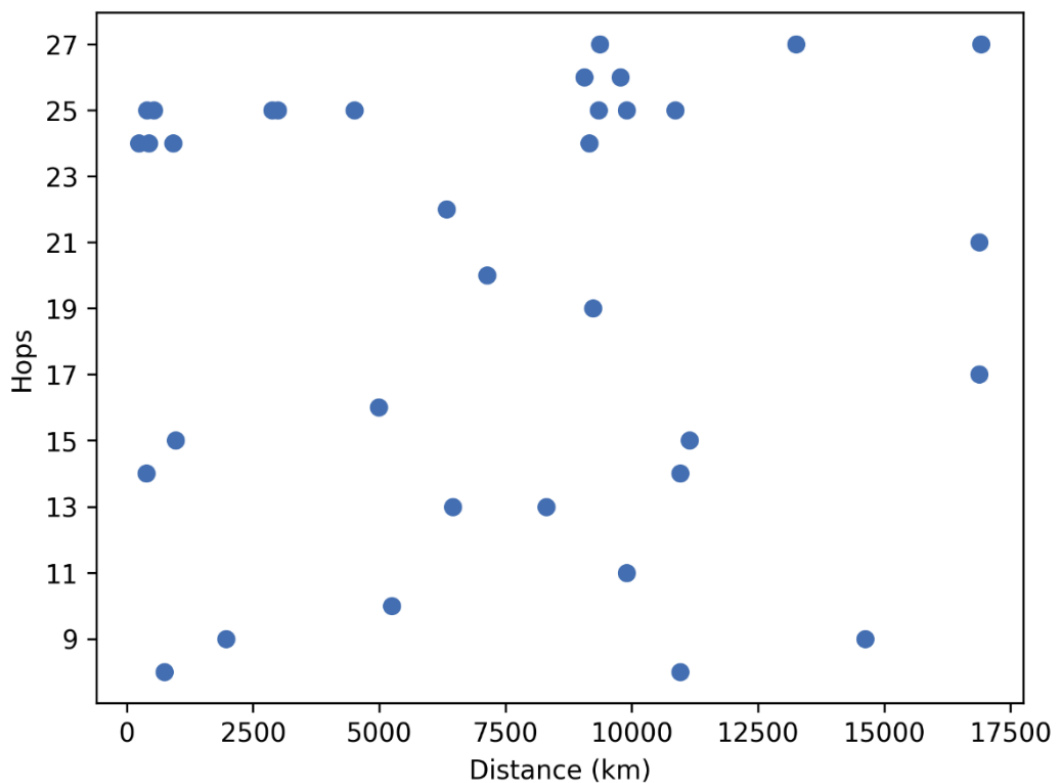


Figure III.8: Graph of correlation between the number of hops and the distance

It can be understood by what we just said earlier. Indeed, we can observe some major axis on the cables network. As with cars, highways enable us to travel a long distance without crossing any intersection. Therefore, we can travel from UK to the USA in one step whereas it could take a few steps to only go from a side of Villeurbanne to another.

As a conclusion, we can say that if this method of measurement is way faster, it is however not relevant at all. Indeed, knowing the number of hops, we cannot know anything about the physical distance nor the actual distance travelled by the packets and therefore on the time. This quantity is completely isolated from the others and doesn't really give us any useful information for the functioning of a network.

4 Measurement of the physical distance and IP geolocation

Finally, we can also measure the physical distance between hosts. Indeed, since the beginning of our project, we have kept comparing every measure we were doing to the distance as crow flies between two devices. It is a quantity we seem to visualize better as we use it on a daily basis, contrarily to the previous ones. However, a question arises : How have we been able since the start of the project to measure the physical distance between two IP addresses ? This question can actually be reduced to the following : How can we geolocate a device with for only information its IP address ?

Actually, at the start of the project, we were simply using the website www.iplocation.net. We had only to give it the IP address of a server and it was giving us the location at a more or less precise scale, using different databases. At the end of the two weeks, we decided to try to understand how these kind of websites were working and to find a method to obtain by ourselves an approximation of the location of an IP.

a Methods of IP geolocation

A first method used to geolocate an IP address is to check to which area corresponds a certain range of IP addresses. Indeed, some ranges of IP addresses are used by a defined place. For example, on the page help.uis.cam.ac.uk, we can find a list of the different IP ranges corresponding to the departments and laboratories of the university Cambridge. For instance, we can find that the range 128.232.0.0/16 is associated to the computer laboratory. The notation /16 means that the subnet of the lab is defined by the 16 first bits of the address, here 128.232. This subnet includes then 65534 host IP addresses that we can create with the 16 last bits.

This way, thanks to databases, we can identify the country, the region or even the city to which belongs an IP address. A map below shows the distribution of some ranges of IP addresses all over the world.

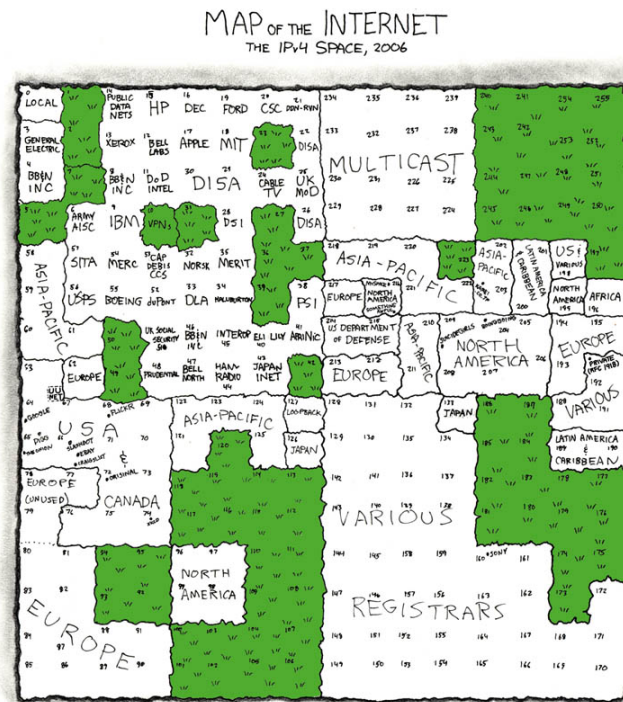


Figure III.9: Map of the internet

Then, another method consists in using giant databases that map IP addresses to their location. There exist different types more or less accurate, more or less updated, and therefore more or less expensive. The main websites providing IP geolocation services use these kind of databases.

During the project, we used `ipinfo`, a Python library, which uses this method, to get the location of a host in longitude and latitude, to quickly locate many hosts at once and graph many things. Using this library is very simple and straightforward:

```
import ipinfo

handler = ipinfo.getHandler(access_token)

async def get_location(url):
    details = handler.getDetails(url)
    result = None
    if hasattr(details, "loc"):
        result = [float(x) for x in details.loc.split(',')]
    return result
```

Computing the distance between two locations was very simple, since we already had the

longitudes and latitudes, so we just had to apply the math.

```
def distance(location_1, location_2):
    lat_1, long_1 = location_1
    lat_2, long_2 = location_2

    dist_lat = math.radians(lat_2 - lat_1)
    dist_long = math.radians(long_2 - long_1)

    a = math.sin(dist_lat / 2) * math.sin(dist_lat / 2) +
        math.cos(math.radians(lat_1)) * math.cos(math.radians(lat_2)) *
        math.sin(dist_long / 2) * math.sin(dist_long / 2)
    dist = 6371 * 2 * math.atan2(math.sqrt(a), math.sqrt(1 - a))

    return dist
```

The website we used in our project allowed us to geolocate an IP address using different databases and we could see that sometimes, they would not agree with each other. Indeed, the accuracy of these databases and of the IP geolocation methods in general is variable and is never 100% accurate. Finding the exact location of an IP is almost impossible but we can find at least an approximation. However, there are some limits to these tools that can make even an approximation very complicated.

b Limits to the IP geolocation

Due to the needs of Internet users, there are some services which aim is, among others, to hide the location of a computer, or more precisely, to hide its IP address that can reveal some information about it.

First, this is the case of the VPN, which makes quite difficult the geolocation of a computer. VPN actually stands for Virtual Private Network, describing very well the functioning of it. Indeed, the VPN, in addition to encrypting all your data, creates a virtual private network with all the users, as if you were at the same place. It gives you a new private IP address and therefore, when you make a request, as in a private network, your request first pass through the VPN which is then going to make the request for you, with its own IP address. The servers receiving your request only see the IP address of the VPN, which induces them into thinking you're at the same location when you are absolutely not. This is what allows some users of VPNs to be able to watch series not available in their countries for example. Indeed, the streaming platform, when it receives a request, tries to locate the computer in order to send it a page adapted to its country. The utilization of VPNs creates a deceptive environment wherein users connected to a VPN in the USA influence the platform to perceive their location as such, consequently delivering content accessible in the USA. This phenomenon underscores the inherent limitation of IP geolocation posed by VPNs.

Another challenge faced by IP geolocation services is the dynamic IP assignments by ISPs. The

latter means Internet Service Provider and as its name indicates, it is charged of connecting a computer to the Internet. For that, it assigns it an IP address but for more flexibility, this address changes each time the device connects to the network. Indeed, each ISP disposes only of a certain number of IP addresses so dynamic IPs allow a more efficient way to use them. Indeed, when a device disconnects, it is not useful to reserve an IP address for it anymore. Instead, it becomes available for another device. Dynamic IPs can also be better from a practical point of view since they don't require manual configuration unlike static-IP. For all these reasons, dynamic IPs are very common and it can represent a challenge for IP geolocation since IPs and locations are very variable, making it more difficult to well update the databases. Also, some IP geolocation databases may not always be up-to-date or accurate. They rely on information by ISPs and other organizations, which may not always be comprehensive or correct.

Moreover, with the increasing use of mobile devices and cellular networks, IP geolocation becomes more challenging due to the dynamic nature of mobile IP assignments and the mobility of users.

c IP geolocation by triangulation

Then, we wanted to try if we could find a way to geolocate a device by ourselves. We chose to proceed with a triangulation method.

For that, we applied a simple protocol:

We chose multiple servers around the world from which we could ping the host we wanted to locate. From this ping, thanks to the figure **III.2**, we were able to find an approximation of the physical distance between each server and the host.

We took the lower bound of the graph of the correlation between physical distance and ping.

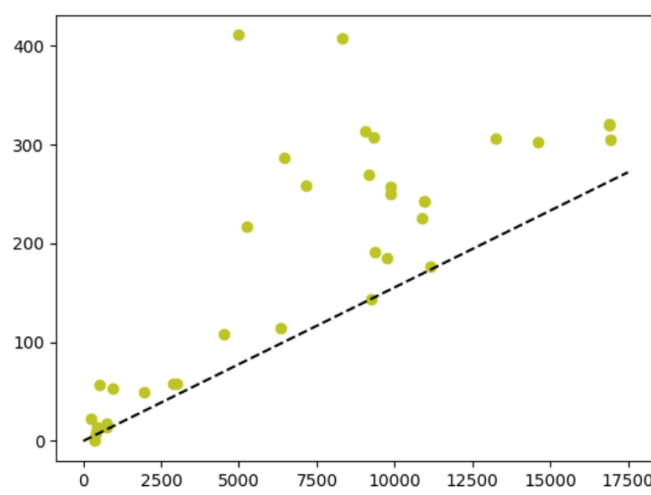


Figure III.10: Approximation of distance to a host

From this, we can tell, with some uncertainty of course, that if the ping from our server to a host is a number y , the distance between the server to the host is at most x .

By tracing circles of a radius equal to this distance x around the different servers, we obtained a zone more or less large on which the circles were overlapping. This zone corresponds to our approximation of the location of the server.

We tried this method multiple times in order to see if it was accurate or not. As we can see in the following graphs, this method gave us good results: the host is in the area located. However, the zone obtained is very large and it was very complicated to obtain good results at first. Indeed, as we said before, the approximation of the physical distance thanks to the ping is quite inaccurate, which made the process harder. We had to make the ping from a lot of servers to reduce the size of the zone and to make it more accurate but it's still not perfect.

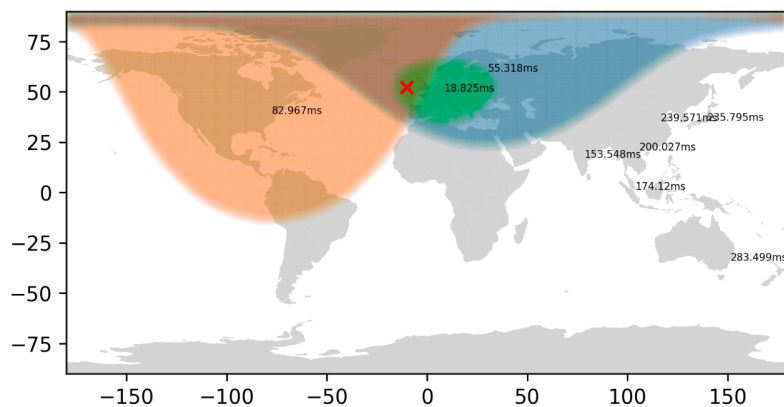


Figure III.11: First example of the geolocation of a host thanks to the triangulation method

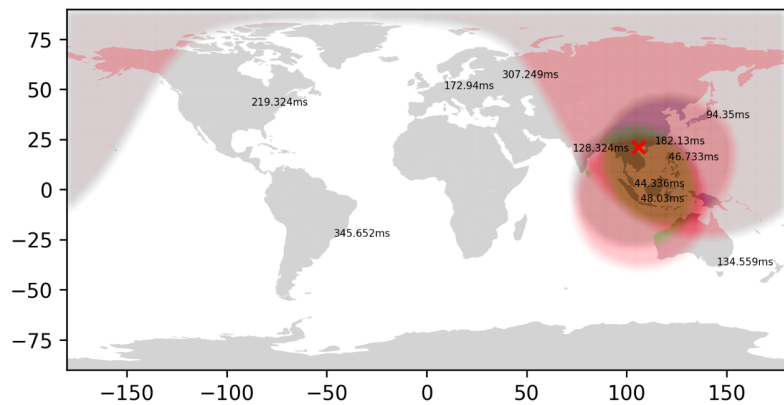


Figure III.12: Second example of the geolocation of a host thanks to the triangulation method

To conclude, even if this method can be done by everyone freely, it is a lot less precise and

reliable than the databases method, which is not perfect either though.

IV Conclusion

To conclude, during this two weeks project, we were able to make a lot of searches and understand better how the Internet works and what it is composed of. Then, we also tried to apprehend the different ways we had to measure “distances” on the Internet : first by measuring the time, then by measuring the real distance travelled by the packets, by counting the number of hops and finally by measuring the physical distance, trying to see how all these values can be linked or not.

We actually saw that a lot of measurements are very complicated and always very inaccurate on the Internet, due to its constant fluctuation. From all our searches, we also arrived to the conclusion that each of the ways of measuring the Internet are not completely linked to each other, they are all a bit independent because there are too many parameters to take into account. Thus, it's very hard to compare them to determine which one is the best way to “measure the Internet”. However, we decided to conclude that it was probably by measuring the time. Indeed, that's the only measurement that really interests us. When we want to connect to another server, we want it to be quick, no matter at what physical distance it is from us. When we want to report a dysfunction of a network, we always refer to a problem of time. That's also why we are still developing new technologies to always lower the propagation time. That is because the most important after all, is the time.

V References

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- [6] [mtrpacket - Python library for pinging](#)
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- [8] [perfops - Allows pinging from many locations in the world](#)